This assignment will not be graded and is only for practice.

1) Consider the following functions

$f_1(x,y)= 5\sqrt{x^2+y^2}$

$f_2(x,y,z) = x^2z^2 + yz + x^2y^2z^2$

$f_3(x,y)= \begin{cases} y\sin\frac{1}{y} + xy & \text{if } y = 0 \\ 0 & \text{if } y = 0. \end{cases}$

If tangent plane exists at the origin for f_i, then enter the value of i.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

2) Which of the following equations represents the tangent plane to the function $f(x,y) = 3x^2 + xy - 2y^2$ at the point (1,-1)

1 point

- ○ $5x + 5y = 0.$
- ○ $5x + 5y - z = 0.$
- ○ $x + y - z = 0.$
- ○ $5x + 5y + z = 0.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$5x + 5y - z = 0.$

3) Consider a function $f(x,y) = \frac{1}{2}(xy - \frac{x}{y})$. Let T be the tangent plane at point (1,2). Which of the following points lie on the tangent plane T?

1 point

- ☐ $(0,0,-\frac{5}{4})$

- [ ] (1,0,$\frac{1}{4}$)
- [ ] (2, -4, $-\frac{9}{4}$)
- [ ] (1, 2, -2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(0,0,$-\frac{5}{4}$)
(2, -4, $-\frac{9}{4}$)

4) Consider a function $f(x, y) = xy$. Let T_1 be the tangent plane at point (1,1), T_2 be the tangent plane at point (1,-1). Then which of the following option is true? ***1 point***

- ○ T_1, T_2 intersect in only one point.
- ○ T_1, T_2 intersect in infinitely many points.
- ○ T_1, T_2 do not intersect.
- ○ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

T_1, T_2 intersect in infinitely many points.

5) Let f be a function from R^2 to R and T be the tangent plane at a point. Then which of the following option(s) is (are) true. ***1 point***

- [ ] T is an affine subspace of the vector space R^3 with respect to usual addition and scalar multiplication.
- [ ] T is never an affine subspace of the vector space R^3 with respect to usual addition and scalar multiplication.
- [ ]

T can be a subspace of the vector space $\mathbb{R}^3$ with respect to usual addition and scalar multiplication.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is an affine subspace of the vector space $\mathbb{R}^3$ with respect to usual addition and scalar multiplication.
T can be a subspace of the vector space $\mathbb{R}^3$ with respect to usual addition and scalar multiplication.

6) Consider a function $f(x, y, z) = e^{x+y} \sin 2z$. Which of the followings is the tangent (hyper) plane at the point $(1, 0, \frac{\pi}{4})$? ***1 point***

○ $u = ex + ey$

○ $u = ex + ey + z$

○ $u = ex + ey + ez$

○ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$u = ex + ey$

7) Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a function and L_f be the linear approximation of f at point (a, b). ***1 point***
Which of the following options is true?

○ L_f is a linear function.

○ L_f is a quadratic function.

○ L_f is a logarithmic function.

- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

L_f is a linear function.

8) Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as $f(x,y) = xy\tan(x)\sin^{-1}(y + \frac{\sqrt{3}}{2}) + 1$. Which of the following is the linear approximation of f at the point $(\frac{\pi}{4}, 0)$ ***1 point***

- [ ] $L_f(x,y) = x + \frac{\pi^2}{12}y$
- [ ] $L_f(x,y) = 1 + \frac{\pi^2}{12}y$
- [ ] $L_f(x,y) = 1 + y$
- [ ] $L_f(x,y) = \frac{\pi^2}{12}y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(x,y) = 1 + \frac{\pi^2}{12}y$

9) Consider the three functions $f_1 = x^2 + y^2, f_2 = xy$ and $f_3 = x + y$. Let T_1, T_2 and T_3 be the tangent planes at points $(-\frac{1}{2}, \frac{1}{2})$ of f_1, (-1, 1,) of f_2 and (0,0) of f_3 respectively. If (a, b, c) are the intersection point of the T_1, T_2 and T_3, then find the value of $a + 2b + c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

10) Which of the following is the linear approximation of the function $f(x,y) = e^x \sin y + xe^x$ at the point (0, 0) ***1 point***

- $L_f(0,0) = 2x + y$
- $L_f(0,0) = x$
- $L_f(0,0) = 2x + 2y$
- $L_f(0,0) = x + y$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(0,0) = x + y$

11) Production of number of breads depends upon the quantity of raw material which includes egg, wheat and sugar . x denotes the number of eggs, y denotes the quantity of wheat (in kg) and z denotes the quantity of sugar (in kg) and defined as $f(x, y, z) = xyz$. Quality of the bread is measured by the value of linear approximation $(L_f(x, y, z))$ with respect to 1 egg, 1 kg wheat, and 1 kg sugar. If value of the linear approximation is greater than 10, then quality of product is not good and if less than 10 then quality of the product is good. ***1 point***

Use this information to answer question.

Which of the following option(s) is (are) true?

- If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.
- If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.
- If 7 eggs, 8 units wheat, 9 units sugar is used to produce breads, then the quality of produced breads is good.
- If 5 eggs, 3 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.
If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

12) Consider a functions $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as: $f(x, y, z) = x^2 + y^2 + z^2$. If $L_f(x, y, z)$ is the linear approximation of $f(x, y, z)$ at point (1, 1, 0), then find the value of $\nabla L_f(x, y, z) \cdot (3, 2, 1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

13) Consider a function $f(x, y, z) = ax + by + cz$, where $a, b, c \in \mathbb{R}$. If $L_f(x, y, z)$ is the linear approximation of $f(x, y, z)$ at the origin, then which of the following options is true? ***1 point***

- $L_f(x, y, z) = 2f(x, y, z)$
- $L_f(x, y, z) = -f(x, y, z)$
- $L_f(x, y, z) = f(x, y, z)$
- $L_f(x, y, z) = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(x, y, z) = f(x, y, z)$

Your score is: 0/13